QIAORONG S. YU

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EDUCATION

2025–2030 **New York University**
Ph.D. in Cognition and Perception (Computational Neuroscience)
Advisors: Prof. David J. Heeger & Prof. Jonathan A. Winawer

2021–2025 **Balliol College, University of Oxford**
Integrated B.A. & M.Sc. in Mathematical and Theoretical Physics
Thesis: *Mental Representations of Spacetime*
Advisor: Prof. Vijay Balasubramanian

PUBLICATIONS

Preprints

Qiaorong, S. Yu, Zhaoze Wang, and Vijay Balasubramanian. "When and Where: A Model Hippocampal Network Unifies Formation of Time Cells and Place Cells." 10.64898/2026.03.22.713480

Peer Reviewed Articles

Qiaorong, S. Yu, Stephanie O'Neil, Xuejian Shen, Mark Vogelsberger, Sownak Bose, Boryana Hadzhyska, Lars Hernquist, Rahul Kannan, Monica Wu, and Ziang Wu. "Measuring the splashback feature: Dependence on halo properties and history." The Open Journal of Astrophysics 8 (2025). 10.33232/001c.146824

PRESENTATIONS

Talks

2024 *The measurements of splashback features of dark matter halos using IllustrisTNG and MillenniumTNG simulations*, MIT Kavli Institute for Astrophysics and Space Research

Posters

2026 *A neural circuit theory with recurrent normalization explains the dynamics of visual working memory*s, The Society for Neuroscience Annual Meeting (SfN) 2026
A Hippocampal CA3 Network Unifies the Formation of Time Cell and Place Cell, Computational System Neuroscience (COSYNE) 2026

2024 *Attempting dream decoding with generalisable visual EEG encoding models*, Cognitive Neuroscience Society (CNS) 2024 Annual Meeting

HONOURS & AWARDS

2026 Dean's Conference Fund ($750), Graduate School of Arts and Sciences, New York University
2025 MacCracken Fellowship, Graduate School of Arts and Sciences, New York University
2024 Sir William Markby Scholarship (£250), Balliol College, University of Oxford
Dr Prosser's Trust Fund (£1000), Balliol College, University of Oxford
JT Hamilton Legacy Fund (£500), Balliol College, University of Oxford
2023 Reynolds Scholarship (£250), Balliol College, University of Oxford
Lubbock Fund (£2000), Balliol College, University of Oxford
2022 Prosser Exhibition (£150), Balliol College, University of Oxford
2020 British Physics Olympiad Round One Top Gold (Global Ranking)
Duke of Edinburgh Award, Silver and Bronze Medals

TEACHING

New York University

2026 Mathematical Tools for Neuroscience I, NEURL-GA / PSYCH-GA, Teaching Assistant
Intro to Psychology, PSYCH-UA, Teaching Assistant

University of Oxford

2022–2023 Comprehensive Oxford Mathematics and Physics Online School (COMPOS) curriculum, Tutor